Drug Repurposing in Rheumatoid Arthritis

Section 4 — working draft (Days 1–3)

4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMPK and NF-κB (Section 3). Insulin resistance is also common in RA and is linked to higher disease activity [Ormseth2013]. It covers metformin and pioglitazone first, then the newer GLP-1 receptor agonists, and finally SGLT2 inhibitors.

4.1.1. Metformin

Metformin is a cheap first-line drug for type 2 diabetes, and it is generally safe [Gharib2021]. Its main mechanism is activation of AMP-activated protein kinase (AMPK), which regulates glucose and lipid metabolism [Son2014]. Metformin also has anti-inflammatory effects [Son2014]. AMPK activation suppresses mammalian target of rapamycin (mTOR), and this changes how T cells behave. For example, metformin inhibits the differentiation of Th17 cells, which are important drivers of chronic inflammation [Son2014].

Animal studies consistently show that metformin reduces arthritis and protects the joint. In the collagen-induced arthritis (CIA) model (Table 1),

immunized mice were given 5 mg/mouse of metformin orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced arthritis severity and restored the Th17/Treg balance: Th17 cells fell and Treg cells rose, with less STAT3 and more STAT5 phosphorylation. In the same model, the drug also reduced osteoclast activity [Son2014]. In rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed that metformin protected the trabecular bone, while histology showed reduced osteoclast formation and synovitis. Metformin also reduced systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [Fan2020].

Human joint tissue also links AMPK to RA activity. In synovium from patients with RA, the active (phosphorylated) form of AMPK-α1 was higher than in synovium from osteoarthritis patients. In their serum, phosphorylated AMPK-α1 rose with the 28-joint Disease Activity Score (DAS28), and total AMPK-α1 rose with TNF-α [Chen2020]. When fibroblast-like synoviocytes (FLS) from these patients were cultured, metformin at high (millimolar) doses slowed cell growth and movement and reduced IL-1β and IL-6 as well as TNF-α [Chen2020]. The authors doubted that such doses can be reached in patients [Chen2020].

Two small clinical studies support adding metformin to

csDMARDs. A randomized, single-blind clinical trial enrolled 66 Egyptian patients with moderately or highly active RA. Of these, 33 patients received metformin 850 mg twice daily in addition to csDMARDs, and the other 33 patients received placebo. This led to reduced C-reactive protein (CRP) levels and DAS28-CRP scores after 6 months compared with placebo, and only three patients withdrew from the metformin group because of gastrointestinal side effects [Gharib2021]. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving csDMARDs only. This was associated with a faster fall in IL-6, TNF-α, HMGB1 and CRP than in the csDMARD-only group [Zhang2023].

Data on RA prevention are conflicting. A large UK cohort followed people recently diagnosed with type 2 diabetes. RA risk was the same in metformin users as in users of other oral diabetes drugs (hazard ratio [HR] after adjustment 1.00, 95% CI 0.82 to 1.22) [Zemedikun2021]. A Mendelian randomization study suggested that metformin may protect against RA, but its genetic markers reflect metformin use and so mostly track diabetes [Liang2023]. Metformin may therefore be a safe and cheap add-on to csDMARDs to lower RA disease activity. Whether it prevents RA is still unclear, and larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on

X-ray are needed.

4.1.2. Pioglitazone

Pioglitazone, a thiazolidinedione, activates the nuclear receptor PPAR-γ (peroxisome proliferator-activated receptor-gamma). Although it is mainly used as an antidiabetic drug, it may also have anti-inflammatory effects. PPAR-γ activation suppresses pro-inflammatory genes and the production of inflammatory cytokines, matrix metalloproteinases (MMPs), and reactive oxygen species (ROS) [Shahin2011]. Another reason to test it in RA is that insulin resistance and inflammation make each other worse [Ormseth2013].

In animals, pioglitazone helped only at high doses or when combined with methotrexate. In the adjuvant-induced arthritis (AIA) rat model, oral pioglitazone reduced disease severity, but only at the top dose of 30 mg/kg/day [Koufany2008]. It reduced bone erosions and inflammation-associated bone loss but failed to prevent cartilage lesions [Koufany2008]. These anti-arthritic doses were also higher than those needed to restore insulin sensitivity, and they increased body fat [Koufany2008]. In a later study using the same model, pioglitazone alone (1.35 mg/kg/day) had no significant effect on disease activity [Roy2017]. Combined with methotrexate (MTX; 0.225 mg/kg/day), it lowered disease activity more than either drug alone ($P < 0.05$) [Roy2017].

Clinical trials show a small benefit at best. An Egyptian

nonrandomized, single-blind trial enrolled 49 patients with RA and type 2 diabetes, all maintained on MTX 15 mg/week [Shahin2011]. The 28 patients with more active disease added oral pioglitazone 30 mg/day for 12 weeks, while the other 21 took placebo [Shahin2011]. At 12 weeks, DAS28 ($P = 0.001$) and CRP ($P < 0.0001$) were lower in the pioglitazone group than in the placebo group, but ESR stayed higher, and weight gain occurred in 25% of treated patients [Shahin2011]. The placebo group also improved from baseline, and the two groups did not start with the same disease activity, which weakens this result [Shahin2011].

In a US crossover trial that was randomized, double-blind and placebo-controlled, 34 patients with RA on stable therapy received pioglitazone 45 mg/day and placebo for 8 weeks each [Ormseth2013]. Pioglitazone produced a modest 9.3% fall in DAS28-CRP ($P = 0.046$) and no significant change in DAS28 using erythrocyte sedimentation rate (ESR; $P = 0.92$) [Ormseth2013]. Lower-extremity edema occurred in 16% of participants while on pioglitazone but in none while on placebo [Ormseth2013].

A larger crossover trial with the same design and dose enrolled 143 non-diabetic patients with RA, with 3 months per treatment arm [Marder2013]. DAS28-CRP decreased by a mean of 0.31 on pioglitazone compared with 0.15 on placebo ($P = 0.02$) [Marder2013]. However, the authors concluded that these findings did not, at that stage, support adding PPAR-γ agonists to the

standard treatment of RA [Marder2013]. The benefit in these trials was small and came mainly from CRP-based scores, not from ESR or joint counts [Ormseth2013][Marder2013]. Given the risk of fluid retention and weight gain, pioglitazone should not be used routinely in RA. It may still be studied as an add-on for patients with RA who also have insulin resistance or type 2 diabetes.

4.1.3. GLP-1 Receptor Agonists

Liraglutide, semaglutide and dulaglutide are receptor agonists of glucagon-like peptide-1 (GLP-1RAs). They raise insulin release when glucose is high, lower glucagon, slow stomach emptying and increase fullness [Israel2026]. Their link to RA comes from two sides. In a cross-sectional study of 151 patients with RA but without diabetes and 210 sex-matched controls, the patients had an impaired incretin-insulin axis, with higher GLP-1, GIP and insulin levels and more insulin resistance [TejeraSegura2017]. GLP-1 receptors are also found on immune cells (T and B cells, monocytes and macrophages) [Israel2026]. There, the drugs have been reported to lower TNF-α, IL-6 and IL-1β, block nuclear factor kappa B (NF-κB) and favor M2 (anti-inflammatory) macrophages [Israel2026].

Clinical data in RA come mainly from a single-center chart review of overweight or obese patients (BMI ≥ 27 kg/m²) [Kellner2025]. It compared 173 patients who took semaglutide (most patients) or tirzepatide (a dual

GIP/GLP-1 receptor agonist) with 42 patients who were prescribed one of these drugs but did not take it [Kellner2025]. Disease activity, graded from clinic notes on a 0–3 scale rather than DAS28, changed by −0.03 versus +0.20 in controls (P = 0.03) [Kellner2025]. This difference is small, and its clinical meaning is unclear. Pain fell by 0.6 cm on a visual analogue scale in the treated group but rose by 1.3 cm in controls (P < 0.001) [Kellner2025]. ESR and CRP did not fall more than in controls, and the pain relief was not linked to weight loss [Kellner2025]. The study was small, and the reasons for starting the drug were not measured [Kellner2025].

Not all data point to a benefit. A case-control study compared 4,535 RA patients with 22,675 controls [Israel2026]. Use of semaglutide or liraglutide in the 10 years before diagnosis was linked to higher odds of new-onset RA (adjusted odds ratios 1.64 and 1.58), and dulaglutide pointed the same way without reaching significance (adjusted odds ratio 1.31) [Israel2026]. This link was seen mainly with short use (≤6 months), and the authors suggest the drugs may cause a short-lived rise in inflammation early in treatment [Israel2026]. Confounding may explain part of this, because obesity and diabetes, the main reasons for giving these drugs, raise RA risk on their own [Israel2026]. The authors add that longer use may lower inflammation through weight loss and better insulin sensitivity [Israel2026]. The two studies also ask

different questions, new-onset RA versus established RA, so their results do not have to conflict.

So far, the evidence is limited, observational and mixed, and we found no randomized trials in RA. GLP-1RAs may help some patients with RA and obesity, but they cannot be recommended yet, and future trials should check whether the first months of treatment briefly worsen inflammation.

4.1.4. SGLT2 Inhibitors

SGLT2 inhibitors are oral drugs for type 2 diabetes; dapagliflozin, for example, blocks sodium–glucose cotransporter 2 (SGLT2) in the kidney, so less glucose goes back into the blood [ElDemerdash2025]. The only animal study in arthritis found clear anti-inflammatory effects. In an Egyptian study (Ain Shams University) of AIA rats induced with complete Freund's adjuvant, three oral daily doses of dapagliflozin (1, 5 and 10 mg/kg) were tested for 3 weeks; the highest dose worked best [ElDemerdash2025]. Compared with untreated arthritic rats, dapagliflozin alone lowered serum NF-κB p65 by 66% and TNF-α by 71% [ElDemerdash2025]. It raised the p-AMPK/t-AMPK ratio in the joint by 374%, while MTX alone did not change it [ElDemerdash2025]. This is the same AMPK pathway described for metformin (Section 4.1.1). It also raised joint gene expression of the autophagy markers ATG-7, Beclin-1 and ULK-1 [ElDemerdash2025]. On X-ray, neither drug alone reduced bone erosion or joint-space narrowing; only the

two drugs together did [ElDemerdash2025].

Human data are limited to RA onset in people with diabetes. A study used TriNetX records to emulate target trials in adults with newly diagnosed type 2 diabetes [Yen2026]. It compared SGLT2 inhibitors with sulfonylureas, dipeptidyl peptidase-4 (DPP-4) inhibitors or pioglitazone (more than 80,000 matched pairs per comparison) [Yen2026]. In the first year, SGLT2 inhibitor use was linked to a slightly lower risk of new RA than sulfonylurea use (0.118% vs 0.131%; HR 0.90, 95% CI 0.84–0.97) [Yen2026]. Against DPP-4 inhibitors (HR 0.91) or pioglitazone (HR 0.92), the difference was not significant [Yen2026]. The gap with sulfonylureas shrank with longer follow-up and lost significance, as it also did in a per-protocol analysis [Yen2026]. The authors point out that one year is short, because RA has a long silent phase before diagnosis [Yen2026].

Unlike GLP-1RAs (Section 4.1.3), SGLT2 inhibitors were not linked to a higher RA risk than other glucose-lowering drugs. Still, we found only one animal study and one observational study, and no trial in patients who already have RA. Their use in RA is therefore still experimental.

4.2. Lipid-Lowering Drugs

Patients with RA have a higher cardiovascular risk, so lipid-lowering drugs are a natural choice to test [Kitas2019]. Statins and fenofibrate also act on immune

cells, so they may help the joints as well as the heart. This section covers both.

4.2.1. Statins

HMG-CoA reductase inhibitors (statins) are widely used in cardiovascular therapy. Statins act mainly by competitively inhibiting the HMG-CoA reductase enzyme, and this reduces de novo cholesterol synthesis [Ren2023]. Statins also have effects other than lowering lipids. For example, they make less isoprenoid, so small GTP-binding proteins such as Ras and Rho are less isoprenylated [Chodick2010][ElBarbary2010]. Statins suppress interferon-γ-induced MHC class II expression, and this reduces T-cell activation [Chodick2010]. This matters in RA, where T-cell activation drives joint inflammation [Xing2015].

Animal studies of statins are mostly positive, but not all. In a preclinical study, mice with serum-induced arthritis were given atorvastatin, pravastatin or simvastatin intravenously (0.2 mg/kg on days 3, 5 and 7). Atorvastatin and pravastatin reduced paw swelling and knee destruction on histopathology, but simvastatin had no significant effect [Walker2017].

As Vandebriel et al. point out, most earlier CIA studies, which often used high intraperitoneal doses, found statins protective [Vandebriel2012]. However, when they gave CIA mice oral atorvastatin at 0.2 mg/day or oral pravastatin at 0.8 mg/day, arthritis started earlier, and every treated mouse became arthritic, against 7 of

12 controls [Vandebriel2012]. They are the only group so far to report harm [Vandebriel2012].

Results in patients are mixed. Two meta-analyses showed that statins reduced disease activity, as measured by DAS28 (standardized mean difference [SMD] −0.55, 95% CI −0.83 to −0.26, and −0.78, 95% CI −1.08 to −0.49) [Xing2015][Ren2023]. However, the included trials were small and mostly conducted at single centers [Xing2015], and both meta-analyses reported high heterogeneity [Xing2015][Ren2023]. In an unblinded randomized study conducted in Egypt, 30 patients with early RA were divided into two groups of 15 and treated for 6 months. Group 1 received conventional RA therapy with MTX and prednisone, while group 2 received MTX + prednisone + atorvastatin. DAS28 was significantly reduced from 6.09 to 3.9 in group 2 compared with 6.19 to 5.27 in group 1 [ElBarbary2010]. In contrast, the largest trial, TRACE RA, a double-blind trial in 3,002 UK patients designed for cardiovascular end points, found no difference in DAS28 between patients who received atorvastatin and those who received placebo [Kitas2019]. Atorvastatin did lower CRP, but the authors judged this fall too small to matter clinically in RA [Kitas2019]. In short, statins lowered RA disease activity in small trials, but the largest trial did not confirm this. For now, statins in RA are used to lower lipids and cardiovascular risk, as in other patients. Trials with disease activity as the main end point are needed

before statins can be used to treat RA itself.

4.2.2. Fenofibrate

Fenofibrate activates PPAR-α and is mainly used to treat high triglycerides [Zhou2012]. In mouse T cells, fenofibrate (5–20 μM) blocked Th17-cell differentiation more as the dose rose, by 74% at 20 μM [Zhou2012]. Other PPAR-α agonists (WY14643, GW7647, and bezafibrate) had no such effect, so fenofibrate may act here through another pathway [Zhou2012]. In AIA rats, fenofibrate (300 mg/kg/day for 12 days) lowered the arthritis score from 11.0 to 5.83, a fall of about 47% ($P < 0.01$) [Castillero2012]. Clinical data on fenofibrate in RA are very limited. No placebo-controlled trial has yet tested whether the effects seen in animals also help patients with RA.

4.3. Angiotensin II Receptor Blockers

The renin–angiotensin system (RAS) regulates blood pressure and fluid balance, and its overactivity is a common cause of hypertension. However, some studies suggest that RAS also contributes to RA, particularly through angiotensin-converting enzyme (ACE), which is overexpressed in the peripheral blood and synovium of patients with RA [Wang2013].

Angiotensin II acts on angiotensin II type 1 receptors (AT1R). This produces ROS and switches on NF-κB, and NF-κB then raises the release of several pro-inflammatory cytokines [Mostafa2020]. In contrast,

angiotensin II type 2 receptors (AT2R) work against AT1R, and their stimulation has been linked to anti-inflammatory and tissue-protective effects [Wang2013].

Animal and cell studies of losartan point in the same direction. In AIA rats, losartan at 10 and 15 mg/kg significantly lowered the polyarthritis index compared with vehicle [Wang2013]. This effect was smaller than that of MTX [Wang2013]. Losartan also shifted the balance between the two receptors, with less AT1R and more AT2R in the spleen and synovium, and this shift tracked the fall in the polyarthritis index [Wang2013]. Another study on AIA rats found that losartan lowered serum IL-6 and TNF-α compared with the untreated arthritic group, together with a rise in serum glutathione (GSH) and a fall in serum malondialdehyde (MDA) [AbdelElGaphar2018]. An in vitro study tested losartan, valsartan, and enalapril on peripheral blood mononuclear cells (PBMCs) collected from 16 patients with RA [Cardoso2018]. Losartan lowered four cytokines (IL-6, IL-17F, IL-22, and IFN-γ), while valsartan and enalapril had few or no effects, so this effect may not be shared by valsartan [Cardoso2018]. However, only one high concentration (100 μM) was tested, and cells from patients with active disease responded less [Cardoso2018].

The main clinical evidence for this class in RA comes from a trial of another angiotensin II receptor blocker (ARB), candesartan. In this Egyptian single-blind,

randomized, placebo-controlled trial, 45 patients with RA (15 per group) were treated for 3 months with add-on candesartan (8 mg/day), atorvastatin (20 mg/day), or placebo [Mostafa2020]. Compared with the control group, candesartan significantly lowered serum IL-1β, TNF-α, and CRP and improved DAS28 and HAQ-DI disability scores, while ESR fell only compared with baseline [Mostafa2020]. Unlike losartan in rats, candesartan did not significantly change MDA or rheumatoid factor (RF), while atorvastatin (Section 4.2.1) lowered both [Mostafa2020]. However, patients were grouped according to their other medical conditions, only 45 of 59 eligible patients were analysed, the groups differed at baseline in several variables, including CRP and HAQ-DI, and follow-up was short [Mostafa2020].

ARBs, particularly losartan, may be a sensible choice for patients with RA who also need treatment for hypertension, although only larger and longer trials of losartan can show this.

4.4. Emerging Candidates

Several other drugs have been proposed for RA, but their evidence is weak, indirect or mixed (Table 2). In rats with CIA, the non-selective β-blocker propranolol reduced inflammation in the heart and made arrhythmias less likely [Lin2020]. Its effect on the joints was small: paw swelling and clinical scores stayed close to those of untreated CIA rats and improved only mildly

late in the disease [Lin2020]. In pooled data from five trials of tocilizumab and/or csDMARDs (5,502 RA patients), the 594 (10.8%) who used β-blockers, mostly β1-selective ones, reached SDAI remission less often (adjusted HR 0.70, 95% CI 0.57–0.87) [Abuhelwa2021].

Data for the psychiatric drugs are weaker still. Sertraline, an SSRI, had no effect on synovitis, cartilage damage or bone erosion in CIA rats, unlike MTX, infliximab and tocilizumab in the same study [Chmielewski2026]. Valproate has been proposed because it inhibits histone deacetylases (HDACs). Two selective inhibitors of one HDAC isoform, HDAC6, lowered arthritis scores in CIA mice [Oh2017], but these inhibitors differ from valproate, and we found no clinical study of valproate in RA.

For DPP-4 inhibitors, the data concern RA risk rather than treatment. A nested case-control study included diabetic patients starting second-line therapy (790 RA cases and 11,850 controls) [Kathe2018]. Compared with other second-line antidiabetic drugs, DPP-4 inhibitor use was not linked to new RA after adjustment (adjusted OR 1.16, 95% CI 0.94–1.43) [Kathe2018]. In patients who did not use insulin, the odds were higher (OR 1.30, 95% CI 1.001–1.985) [Kathe2018]. We found no RA-specific study of allopurinol. For now, these drugs remain hypotheses or have mixed results, and better animal studies should come before any trial in patients with RA.